

1. How can data be useful? <https://www.youtube.com/watch?v=8xniRSjRyCQ>
2. Nearest Neighbour: <https://www.youtube.com/watch?v=47jEgvtKHuk>
3. Representation: <https://www.youtube.com/watch?v=5O3joUSYeYk>
4. Linear Classifier: <https://www.youtube.com/watch?v=P92mkhzt6Hg>
5. Performance Metrics - I: https://www.youtube.com/watch?v=_eKO3j5W2nQ
6. Performance Metrics - II: <https://www.youtube.com/watch?v=dG0A2U8B8zQ>
7. Curse of Dimensionality: <https://www.youtube.com/watch?v=w5phdIBr3Sw>
8. Data from multivariate distribution:
https://www.youtube.com/watch?v=AsQORY4_H9Q
9. Representation - II: <https://www.youtube.com/watch?v=3WsDIkP8kMo>
10. Data as matrix: https://www.youtube.com/watch?v=VBfo4L-Y_ug
11. Feature Selection and Extraction:
<https://www.youtube.com/watch?v=DpXBPHnAmZw>
12. Distances and Similarities: <https://www.youtube.com/watch?v=m46Exp1E6X8>
13. Supervised Learning formulation: <https://www.youtube.com/watch?v=v9DKrX-3gEA>
14. Linear Regression: <https://www.youtube.com/watch?v=SSuYxDOxANI>
15. Regularization in Regression: <https://youtu.be/DnRWPq1UGyA>
16. MLE: <https://youtu.be/p8XWxnBzUbk>
17. Naive Bayes: <https://youtu.be/8FZjrckX52E>
18. MSE as MLE: <https://youtu.be/XfmZ3Ap-xHI>
19. MSE using GD: <https://youtu.be/i08JdWDc1MY>
20. Geometry of gaussians: <https://youtu.be/DUOJM728TT4>
21. Orthogonal Line Fitting: <https://youtu.be/likh0NUdvnc>
22. PCA: <https://youtu.be/YpLantxHNJ8>
23. PCA as compression: <https://youtu.be/EQfxYZiE0tg>
24. Eigen Faces: <https://youtu.be/X1cElkXjewI>
25. Eigen Decomposition: https://youtu.be/___EBdS7pvo0
26. Optimization and Eigen Vectors: <https://youtu.be/GvvRuEypvfw>
27. SVD: <https://youtu.be/oN7ddlkzcic>
28. LSI: <https://youtu.be/rMaDnRLZAzM>
29. Rank and Recommendations: <https://youtu.be/tP9e0PlmMKo>
30. LDA-1: https://youtu.be/6yKc_VdQ6Ys
31. LDA-2: https://youtu.be/1gsL_KYaN1o
32. Generalisation and Overfitting: <https://youtu.be/GjSXRaurxi0>
33. Occam's razor: <https://youtu.be/nwd0h8SkSAo>
34. Validation: <https://youtu.be/pvQNBPCgg8>
35. Bias and Variance: <https://youtu.be/lqdY9NWdhLk>

- 36. Loss Functions 1 : <https://youtu.be/ZIKVS7RKFXg>
- 37. Introduction to Pytorch - <https://www.youtube.com/watch?v=u3rUkkh-Rac>
- 38. Neural Networks Basics - <https://www.youtube.com/watch?v=xnJuNxURtik>
- 39. SVM 1 - https://www.youtube.com/watch?v=9_DJ4KvyYoo
- 40. SVM 2 - <https://www.youtube.com/watch?v=ffF8UnbheLk>